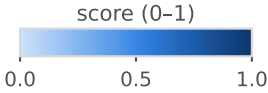


MemVal capacity profile of the model zoo

Cell colour and number: capacity score. In-cell strip: coverage, the share of the capacity's dimension weight that was measurable for that arm. Score and coverage are two claims and are never multiplied.



A Capacity score per arm

	Continual retention	One-shot learning	Pattern completion	Sequence disambiguation	Serial order
AHN rate · one matrix	0.97 coverage 0.90	0.87 coverage 0.75	0.64	0.61	0.68 coverage 0.30
theta rate · one matrix	0.97 coverage 0.90	0.90	0.63	0.61	0.68 coverage 0.30
tPC rate · all-plastic	0.81 coverage 0.90	0.37 coverage 0.75	0.81	0.49	0.73 coverage 0.30
recirc rate · all-plastic	0.78 coverage 0.90	0.46 coverage 0.75	0.69 coverage 0.75	0.51	0.73 coverage 0.50
EP rate · all-plastic	0.75	0.47	0.60	0.56	0.65 coverage 0.50
DTS-ESN rate · readout only	0.74 coverage 0.80	0.88 coverage 0.75	0.71	0.68	0.88 coverage 0.80
spiking EP spiking · all-plastic	0.37 coverage 0.60	0.15 coverage 0.75	0.04 coverage 0.75	0.26	0.48 coverage 0.50
Vieth STDP spiking · one matrix	0.45 coverage 0.60	0.00 coverage 0.75	0.00 coverage 0.75	0.27	0.50 coverage 0.50

B Dimension score per arm, with the reason a dimension was not scored

	Continual retention					One-shot learning			Pattern completion			Sequence disambiguation		Serial order				
	load	plasticity under load	contingency	relevance	schema consistency	presentations	schema consistency	streamed	cue corruption	cue point	cue completeness	contextual overlap	item similarity	unrolling	binding ordinal	establishment	interval retention	interval generation
AHN	1.00	1.00	0.91		1.00	1.00	0.61		0.75	0.50	0.67	0.71	0.47	0.52		1.00		
theta	1.00	1.00	0.91		1.00	1.00	0.61	1.00	0.75	0.45	0.67	0.71	0.47	0.52		1.00		
tPC	0.78	1.00	0.78	0.61		0.44		0.24	0.74	1.00	0.64	0.43	0.58	0.60		1.00		
recirc	0.71	1.00	0.71	0.81		0.48		0.43	0.42	1.00		0.55	0.45	0.50	0.94	0.75		
EP	0.37	1.00	0.88	0.94	0.57	0.46	0.42	0.56	0.27	0.82	0.82	0.55	0.57	0.41	0.73	0.99		
DTS-ESN	1.00	1.00	0.40			0.90		0.83	0.46	0.91	0.83	0.87	0.39	0.57		1.00	0.98	0.99
spiking EP	0.37		0.37			0.14		0.17	0.08	0.00		0.13	0.45	0.26	0.78	0.32		
Vieth STDP	0.27		0.58			0.00		0.00	0.00	0.00		0.15	0.45	0.34	0.74	0.32		
	w 0.25	w 0.20	w 0.35	w 0.10	w 0.10	w 0.50	w 0.25	w 0.25	w 0.40	w 0.35	w 0.25	w 0.60	w 0.40	w 0.20	w 0.20	w 0.10	w 0.25	w 0.25

guard failed: section had no dynamic range for this arm not applicable: arm lacks the declared capability